

Polymerisation

A Level Chemistry

Condensation polymerisation

In **condensation polymerisation** 缩合聚合, monomers join into a long chain **and** a small molecule (such as water or HCl) is lost each time a new bond forms. Each monomer needs **two** reactive groups, so the chain can grow at both ends.

Polyesters

A **polyester** 聚酯 has many ester links along its chain. You can make one from:

- a **diol** 二醇 (two -OH groups) and a **dicarboxylic acid** 二羧酸 (two -COOH groups), or a diol chloride.
- a single hydroxycarboxylic acid, which has both an -OH and a -COOH .

Polyamides

A **polyamide** 聚酰胺 has many amide links. You can make one from:

- a **diamine** 二胺 (two -NH_2 groups) and a dicarboxylic acid or a diol chloride.
- a single aminocarboxylic acid, or from **amino acids** 氨基酸 joining together (proteins are natural polyamides).

Repeat units and monomers

The **repeat unit** 重复单元 of a condensation polymer contains parts of **both** monomers, minus the atoms lost as the small molecule. To find the **monomers** 单体 from a section of polymer, break the chain at each ester or amide link, then add back -OH and -H (or -Cl).

Predicting the type of polymerisation

Clue	Type
the monomer has a $\text{C}=\text{C}$ double bond, and nothing else is lost	addition polymerisation 加成聚合
each monomer has two functional groups, and a small molecule is lost; the chain has ester or amide links	condensation polymerisation

Degradable polymers

- **poly(alkene)s** 聚烯烃 are chemically inert —they have only strong, non-polar C-C and C-H bonds, so they are hard to **biodegrade** 可生物降解 and last a long time.
- some polymers are made so that light can break them down (they are photodegradable).
- polyesters and polyamides **are** biodegradable, because their ester and amide links can be broken by acidic or alkaline **hydrolysis** 水解.